Polar Equations and Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 6.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Recognizing polar graphs

Identify the graph of each polar equation:

a $r = 3$

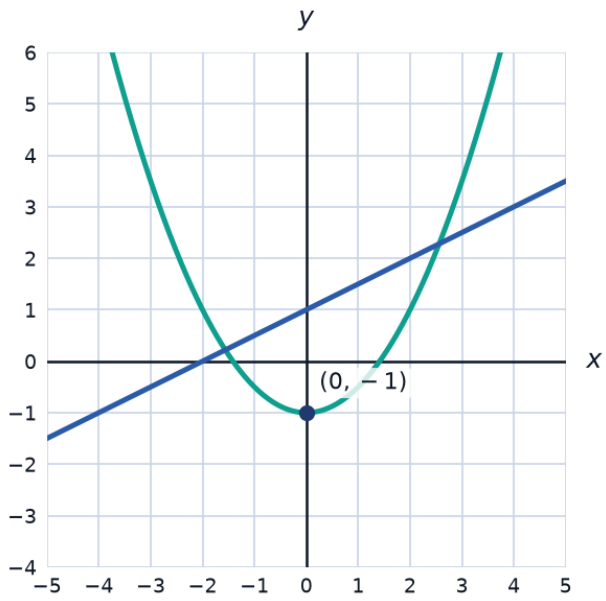
b $\theta = \frac{\pi}{6}$

c $r = 2 + 2\cos\theta$

d $r = 4\sin 3\theta$

2 [Foundational] • Target: Recognizing polar graphs

For the limaçon $r = 1 + 2\cos\theta$:



a Explain how you know it has an inner loop.

b Find the values of θ in $[0, 2\pi)$ at which the curve passes through the pole.

3 [Foundational] • Target: Recognizing polar graphs

For the rose $r = 5\sin 2\theta$:

a State the number of petals.

b State the maximum value of $|r|$ and the smallest positive θ at which it occurs.

4 [Foundational] • Target: Recognizing polar graphs

Determine whether the graph of $r = 2 + \sin\theta$ passes through the pole, and state the type of limaçon.

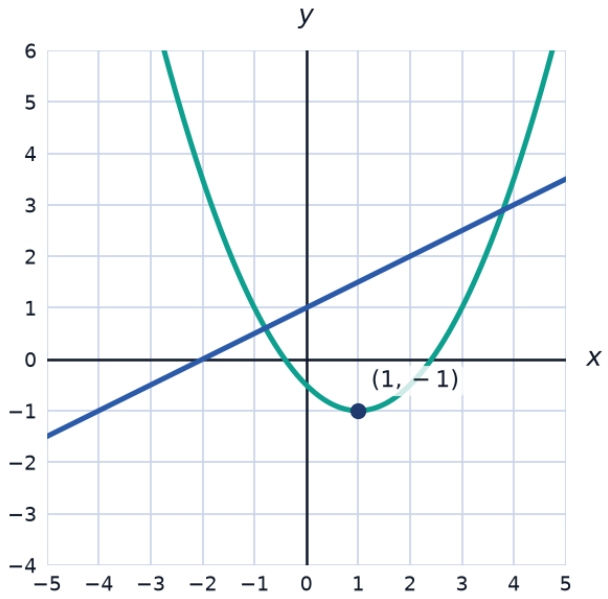
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5 [Foundational] • Target: Symmetry tests

Test $r = 3\cos 2\theta$ for symmetry about the polar axis (replace θ with $-\theta$) and about the pole (replace r with $-r$).

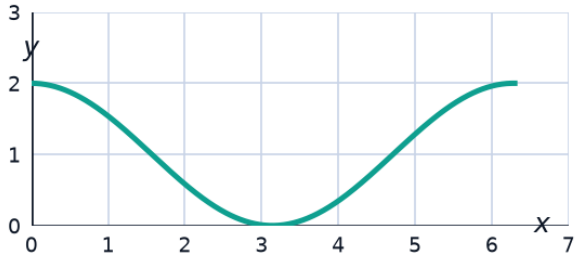
6 [Foundational] • Target: Symmetry tests

Test $r = 2 + 2\sin\theta$ for symmetry about the line $\theta = \frac{\pi}{2}$ (replace θ with $\pi - \theta$).



7 [Foundational] • Target: Graphing from a table of values

For the cardioid $r = 1 + \cos\theta$, complete the table of r-values: $\theta = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$.



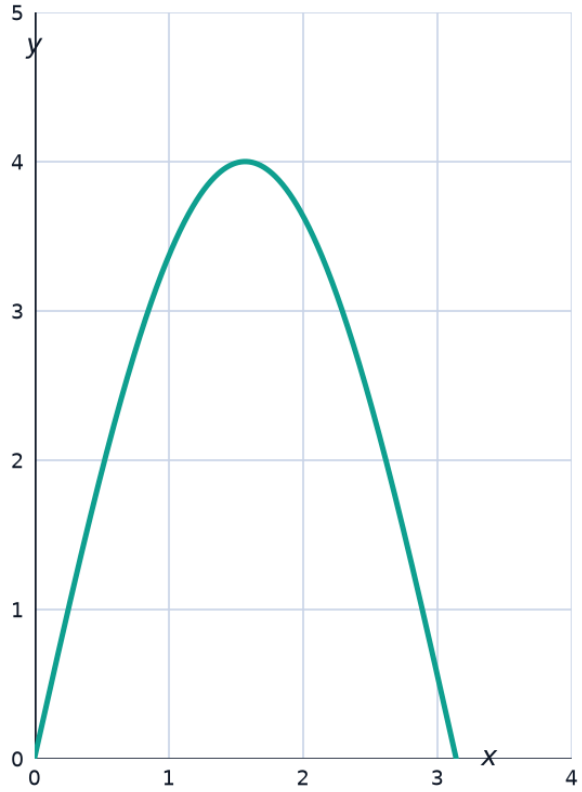
a $r(0)$ **b** $r\left(\frac{\pi}{2}\right)$ **c** $r(\pi)$ **d** $r\left(\frac{3\pi}{2}\right)$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

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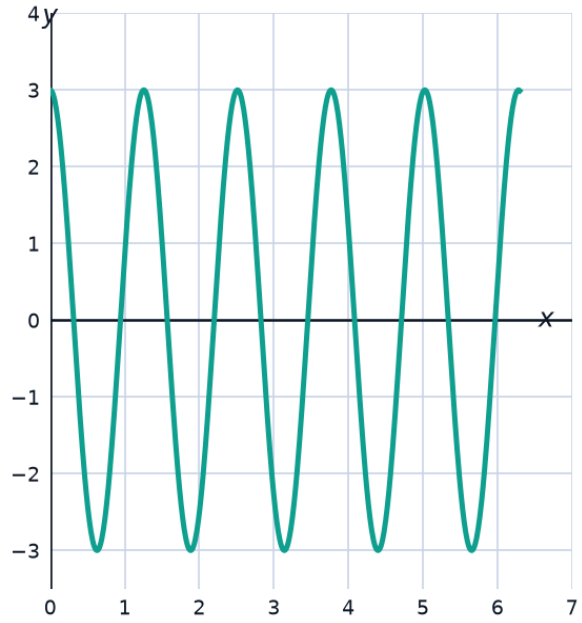
8 [Intermediate] • Target: Graphing from a table of values

For the circle $r = 4\sin\theta$, find the value(s) of θ in $[0, \pi)$ where r reaches its maximum, and state that maximum value.



9 [Intermediate] • Target: Rose curves in depth

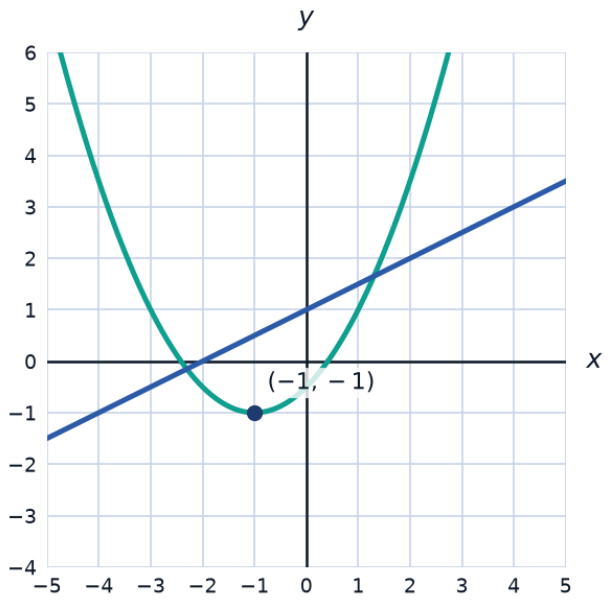
For the rose $r = 3\cos 5\theta$, state the number of petals and the length of each petal (maximum $|r|$).



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10 **[Intermediate]** • **Target: Rose curves in depth**

Explain, using the general rule for rose curves $r = a\cos(n\theta)$ or $r = a\sin(n\theta)$, why n odd gives n petals but n even gives $2n$ petals.



11 **[Intermediate]** • **Target: Identifying limaçon subtypes**

Classify each limaçon $r = a + b\cos\theta$ as having an inner loop, being dimpled, or being convex (roughly oval, no dimple), based on the ratio $\frac{a}{b}$:

a $r = 1 + 3\cos\theta$

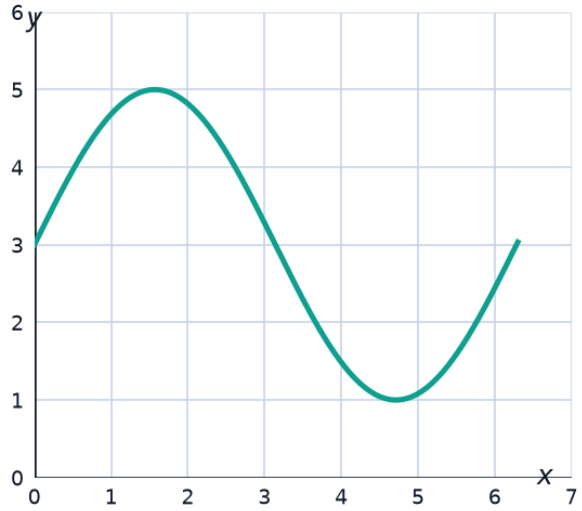
b $r = 3 + 2\cos\theta$

c $r = 4 + \cos\theta$

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12 **[Intermediate]** • **Target: Finding maximum and minimum r-values**

For the limaçon $r = 3 + 2\sin\theta$, find the maximum and minimum values of r, and the angles at which they occur.



13 **[Intermediate]** • **Target: Comparing sine- and cosine-based polar curves**

Explain how the graph of $r = 3 + 3\sin\theta$ compares to $r = 3 + 3\cos\theta$ (same cardioid shape, different orientation).

14 **[Intermediate]** • **Target: Sketching from key features**

Without fully graphing, list the key features you would compute first to sketch $r = 2 - 4\cos\theta$ (max/min r, pole crossings, symmetry).

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

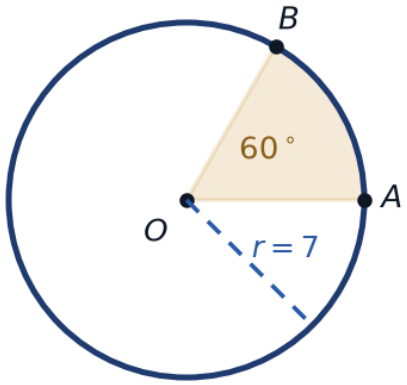
15 **[Challenging]** • **Target: A capstone graphing problem** • **Cross-Lesson: 2.02 Radian Measure and Arc Length**

For the rose $r = 4\cos(3\theta)$, find: (a) the number of petals, (b) the maximum value of r, (c) the value(s) of $\theta \in [0, \pi)$ where the first petal reaches its maximum length, and (d) the angles where the curve passes through the pole in $[0, \pi)$.

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16 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Without graphing, state whether each polar equation is a circle, a line, a rose, or a limaçon, based purely on its algebraic form: $r = 7$, $r = 5\sin(4\theta)$, $\theta = \frac{2\pi}{5}$, $r = 2 - 2\cos\theta$.



17 [Challenging] • Target: Graphing from a table of values • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Two lines have slopes $m_1 = -1$, $m_2 = -1$ and y-intercepts $b_1 = -2$, $b_2 = -2$. Classify the system.

18 [Challenging] • Target: Rose curves in depth • Cross-Lesson: 3.01 The Unit Circle and Exact Values

For $f(x) = 2x^3 + 2x^2 - x + 6$, which values are possible rational roots (by the Rational Root Theorem)?

19 [Challenging] • Target: Identifying limaçon subtypes • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

Find the constant term (independent of x) in the binomial expansion of $\left(2x^2 - \frac{1}{x}\right)^9$.

20 [Challenging] • Target: Finding maximum and minimum r-values • Cross-Lesson: 3.03 Sum and Difference Formulas

Find $\lim_{x \to 25} \frac{\sqrt{x} - 5}{x - 25}$. (attempt 3)

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Circles and Lines in Polar Form

What is the graph of $r = a$ where a is a constant?

a. A circle centered at the origin with radius $|a|$

b. A line through the origin

c. A parabola

d. An ellipse

22 [Foundational] • Target: Circles and Lines in Polar Form

What is the graph of $\theta = \alpha$ where α is a constant angle?

a. A straight line passing through the origin with slope $\tan \alpha$

b. A circle

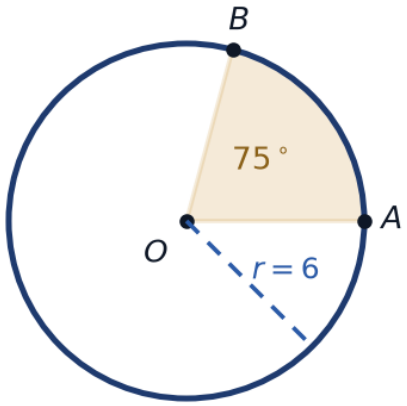
c. A ray only

d. A spiral

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23 [Foundational] • Target: Circles and Lines in Polar Form

Describe the circle given by $r = -6\sin\theta$.



a. Centered at $(0, -3)$ with radius 3

b. Centered at $(0, 3)$ with radius 3

c. Centered at $(0, 0)$ with radius 6

d. Centered at $(-3, 0)$ with radius 3

24 [Foundational] • Target: Limaçons and Cardiods

Under what condition does the limaçon $r = a + b\cos\theta$ $(a, b > 0)$ form a cardioid (heart-shaped curve)?

a. $a/b = 1$ (i.e. $a = b$)

b. $a/b < 1$

c. $1 < a/b < 2$

d. $a/b \geq 2$

25 [Foundational] • Target: Limaçons and Cardiods

Which equation produces a limaçon with an inner loop?

a. $r = 2 + 3\cos\theta$

b. $r = 3 + 3\cos\theta$

c. $r = 4 + 2\cos\theta$

d. $r = 3 + 2\cos\theta$

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26 [Foundational] • Target: Limaçons and Cardiods

What type of curve is $r = 5 + 2\sin\theta$?

a. A convex (dimple-free) limaçon

b. A cardioid

c. A limaçon with an inner loop

d. A dimpled limaçon

27 [Foundational] • Target: Rose Curves

How many petals does the rose curve $r = 4\cos(3\theta)$ have?

a. 3 petals

b. 6 petals

c. 4 petals

d. 12 petals

28 [Foundational] • Target: Rose Curves

How many petals does the rose curve $r = 2\sin(4\theta)$ have?

a. 8 petals

b. 4 petals

c. 16 petals

d. 2 petals

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Rose Curves

What is the maximum length of each petal in $r = 5\cos(2\theta)$?

a. 5

b. 10

c. 2.5

d. 2

30 [Intermediate] • Target: Lemniscates and Spirals

What figure-eight-shaped curve is given by $r^2 = 9\cos(2\theta)$?

a. A lemniscate

b. A cardioid

c. A rose curve

d. A spiral of Archimedes

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31 [Intermediate] • Target: Lemniscates and Spirals

What type of curve is represented by the polar equation $r = 2\theta$ for $\theta \geq 0$?

a. Spiral of Archimedes

b. Logarithmic spiral

c. Lemniscate

d. Circle

32 [Intermediate] • Target: Lemniscates and Spirals

Which symmetry test verifies that a polar graph is symmetric with respect to the polar axis (the horizontal axis)?

a. Replacing (r, θ) with $(r, -\theta)$ yields an equivalent equation

b. Replacing (r, θ) with $(-r, \theta)$

c. Replacing θ with $\theta + \pi$

d. Replacing r with $1/r$

33 [Intermediate] • Target: Rose Curves

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \leq \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

34 [Intermediate] • Target: Lemniscates and Spirals

What is the Cartesian equation of the polar curve $r = 5$?

a. x+y=25

b. $x^2 - y^2 = 25$

c. $x^2 + y^2 = 5$

d. $x^2 + y^2 = 25$

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35 [Intermediate] • Target: Recognizing polar graphs

What is the Cartesian equation of the polar curve $r = 6$?

a. $x^2 + y^2 = 36$

b. x+y=36

c. $x^2 + y^2 = 6$

d. $x^2 - y^2 = 36$

36 [Intermediate] • Target: Symmetry tests

What is the Cartesian equation of the polar curve $r = 7$?

a. x+y=49

b. $x^2 - y^2 = 49$

c. $x^2 + y^2 = 7$

d. $x^2 + y^2 = 49$

37 [Intermediate] • Target: Graphing from a table of values

What is the Cartesian equation of the polar curve $r = 8$?

a. x+y=64

b. $x^2 + y^2 = 8$

c. $x^2 - y^2 = 64$

d. $x^2 + y^2 = 64$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Rose curves in depth • Cross-Lesson: 3.01 The Unit Circle and Exact Values

What is the Cartesian equation of the polar curve $r = 9$?

a. $x^2 + y^2 = 81$

b. $x^2 - y^2 = 81$

c. x+y=81

d. $x^2 + y^2 = 9$

39 [Challenging] • Target: Identifying limaçon subtypes • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

What is the Cartesian equation of the polar curve $r = 1$?

a. $x^2 + y^2 = 1$

b. $x^2 - y^2 = 1$

c. $x^2 + y^2 = 2$

d. x+y=1

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40 [Challenging] • Target: Finding maximum and minimum r-values • Cross-Lesson: 3.03 Sum and Difference Formulas

What is the Cartesian equation of the polar curve $r = 2$?

a. $x^2 + y^2 = 2$

b. $x^2 + y^2 = 4$

c. x+y=4

d. $x^2 - y^2 = 4$

41 [Challenging] • Target: Comparing sine- and cosine-based polar curves • Cross-Lesson: 3.04 Double-Angle Formulas and Trigonometric Equations

What is the Cartesian equation of the polar curve $r = 3$?

a. $x^2 - y^2 = 9$

b. x+y=9

c. $x^2 + y^2 = 9$

d. $x^2 + y^2 = 3$

42 [Challenging] • Target: Sketching from key features • Cross-Lesson: 4.01 Circles and Parabolas

What are the maximum and minimum values of $y = 2\sin(x) - 2$?

a. Max 1, Min -5

b. Max 0, Min -4

c. Max -2, Min 2

d. Max 2, Min -2

43 [Challenging] • Target: A capstone graphing problem • Cross-Lesson: 4.02 Ellipses

What are the maximum and minimum values of $y = 3\sin(x) + 1$?

a. Max 3, Min -3

b. Max 4, Min -2

c. Max 1, Min -1

d. Max 5, Min -3

44 [Challenging] • Target: Distinguishing curve families by equation form alone • Cross-Lesson: 4.03 Hyperbolas

What are the maximum and minimum values of $y = 5\sin(x) + 1$?

a. Max 1, Min -1

b. Max 7, Min -5

c. Max 5, Min -5

d. Max 6, Min -4

45 [Challenging] • Target: Circles and Lines in Polar Form • Cross-Lesson: 5.01 Vectors: Components and Operations

What are the maximum and minimum values of $y = 4\sin(x) + 1$?

a. Max 5, Min -3

b. Max 1, Min -1

c. Max 4, Min -4

d. Max 6, Min -4

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